

A Krawtchouk XX Chain: Exact Propagation, Perfect Mirror Transfer, and the Full Fermionic Phase

HCS-C366 source-local theorem package

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Abstract

We identify the engineered open XX chain with the spin- $N/2$ representation of J_x , obtaining its complete Krawtchouk spectrum and propagator for every chain length. An endpoint excitation follows an exact binomial law and is mirrored perfectly at time π/Ω . Fermionic second quantization then gives the propagator in every excitation sector and exposes the wedge-reordering sign that a one-particle argument cannot determine. Finite exact ledgers audit the formulas; representation theory proves the all-size statements.

1 Frozen chain and one-particle spectrum

On the ordered basis $\{|j\rangle : 0 \leq j \leq N\}$, let

$$h_N = \frac{\Omega}{2} \sum_{j=0}^{N-1} \sqrt{(j+1)(N-j)} (|j\rangle\langle j+1| + |j+1\rangle\langle j|), \quad \Omega > 0. \quad (1)$$

Identifying $|j\rangle$ with the J_z -weight $j - N/2$, the standard matrix element of J_x makes (1) exactly ΩJ_x .

Theorem 1 (Krawtchouk spectral resolution). *The simple eigenvalues are*

$$E_r = \Omega(N/2 - r), \quad 0 \leq r \leq N. \quad (2)$$

If

$$K_r(j) = \sum_{\ell} (-1)^{\ell} \binom{j}{\ell} \binom{N-j}{r-\ell}, \quad (3)$$

then normalized eigenvectors have components

$$v_r(j) = \frac{\sqrt{\binom{N}{j}} K_r(j)}{\sqrt{2^N \binom{N}{r}}}. \quad (4)$$

Proof. Spin rotation sends the J_z weights to the J_x weights, proving (2). Pascal identities in the tridiagonal recurrence prove $h_N v_r = E_r v_r$. Coefficient extraction from

$$\sum_j \binom{N}{j} (1-z)^j (1+z)^{N-j} (1-w)^j (1+w)^{N-j} = 2^N (1+zw)^N$$

gives $\sum_j \binom{N}{j} K_r(j) K_s(j) = 2^N \binom{N}{r} \delta_{rs}$. Thus the $N+1$ displayed vectors are a complete orthonormal basis. QED

Theorem 2 (Exact endpoint propagation). *For $U_1(t) = e^{-it h_N}$,*

$$\langle k | U_1(t) | 0 \rangle = (-i)^k \sqrt{\binom{N}{k}} \sin^k \frac{\Omega t}{2} \cos^{N-k} \frac{\Omega t}{2}. \quad (5)$$

At $t_* = \pi/\Omega$, $U_1(t_*) = (-i)^N R$, where $R|j\rangle = |N-j\rangle$.

Proof. The spin- $N/2$ module is the symmetric power of N spin halves. Expanding $e^{-it\Omega\sigma_x/2}|0\rangle = \cos(\Omega t/2)|0\rangle - i \sin(\Omega t/2)|1\rangle$ in normalized symmetric weights gives (5). At angle π , every factor flips with phase $-i$. QED

This identity is the *single-particle propagator owner*. In particular, at $t_*/2$ the endpoint-start probabilities are $2^{-N} \binom{N}{k}$.

2 The many-body theorem

Let $H_N = d\Gamma(h_N)$, and order every fermionic basis wedge increasingly.

Theorem 3 (Fock-space mirror and complete energies). *On the m -particle sector,*

$$e^{-itH_N} = \bigwedge^m U_1(t). \quad (6)$$

At t_* ,

$$|j_1 < \cdots < j_m\rangle \mapsto (-i)^{mN} (-1)^{m(m-1)/2} |N - j_m < \cdots < N - j_1\rangle. \quad (7)$$

For $S \subseteq \{0, \dots, N\}$, $|S| = m$, the corresponding energy is

$$E_S = \Omega \left(mN/2 - \sum_{r \in S} r \right), \quad (8)$$

and all degeneracies are encoded by

$$\prod_{r=0}^N (1 + yq^r) = \sum_{m=0}^{N+1} y^m q^{m(m-1)/2} \mathcal{G}_{N+1,m}(q). \quad (9)$$

Here the Gaussian q -binomial polynomial is fixed by

$$\mathcal{G}_{n,m}(q) = \mathcal{G}_{n-1,m}(q) + q^{n-m} \mathcal{G}_{n-1,m-1}(q), \quad \mathcal{G}_{n,0}(q) = \mathcal{G}_{n,n}(q) = 1, \quad (10)$$

with out-of-range entries zero.

Proof. A number-conserving quadratic fermion Hamiltonian induces the exterior power of its one-particle unitary. At t_* , each occupied orbital contributes $(-i)^N$. Reflection reverses all m wedge factors, and restoring increasing order needs $m(m-1)/2$ transpositions. Exterior products of the one-particle eigenbasis prove (8).

Let $A_{N,m}(q)$ be the coefficient of y^m in the product in (9). Splitting subsets according to whether they contain N gives $A_{N,m} = A_{N-1,m} + q^N A_{N-1,m-1}$. The expression $q^{m(m-1)/2} \mathcal{G}_{N+1,m}(q)$ satisfies the same recurrence: the second term of (10) supplies q^{N+1-m} , and shifting the prefactor from m to $m-1$ supplies q^{m-1} . The empty product is the common base case, so induction proves (9) coefficient by coefficient. QED

Equation (7) is the *many-body phase owner*; deleting its reordering sign changes the unitary in entire excitation sectors.

Source and claim boundary

Christandl, Datta, Ekert, and Landahl introduced fixed-coupling perfect transfer networks in *Physical Review Letters* 92 (2004), 187902, doi:10.1103/PhysRevLett.92.187902. Albanese, Christandl, Datta, and Ekert treated mirror inversion in *Physical Review Letters* 93 (2004), 230502, doi:10.1103/PhysRevLett.93.230502. They establish lineage; the frozen formulas above are rederived, and no literature-priority claim is made.